Isolation bed capacity for Bundibugyo Ebola in Nairobi and Mombasa: a two-city modelling study

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Working manuscript, first draft. All quantitative results are outputs of a deterministic and stochastic transmission model with assumed parameters; they are planning scenarios, not forecasts.

Abstract

Background. In October 2026 Kenya confirmed its first imported cases of Bundibugyo virus disease (BVD) during the largest Ebola outbreak on record in the Democratic Republic of the Congo. Designated isolation capacity is heavily concentrated in Nairobi: of roughly 241 beds nationally, Mombasa — a port city of 1.2 million — has four. We asked how many isolation beds Mombasa would need, and how that number depends on how fast an outbreak is detected.

Methods. We built a two-group (formal and informal-settlement) SEIHFR transmission model for Nairobi and Mombasa, parameterised from the Bundibugyo and Ebola literature and 2019 census data, in which admission to isolation is capped by bed capacity. We computed the basic reproduction number by the next-generation matrix, the probability that two imported cases start a self-sustaining outbreak using a negative-binomial branching process, and the isolation beds required for containment using a stochastic tau-leap simulation. We analysed two bifurcations: one in bed capacity and one in the surveillance-to-response delay.

Results. At a reference R_0 of 2.0 the model gives R_0 = 1.95 for Nairobi and 2.18 for Mombasa, rising above 4 once isolation wards are full. From two index cases the probability of a self-sustaining outbreak is 32–36% under strong Ebola overdispersion (k = 0.18). With a response beginning seven days after the first isolation admission and a 60% reduction in community contacts, Mombasa needs about 22 total isolation beds (18 at Utange, in addition to the four at Port Reitz) for a 95% probability of containment, and about 36 for 99%; Nairobi's 57 beds are more than sufficient. The requirement is highly sensitive to detection speed: it roughly triples if detection is delayed from 3 to 21 days. The bed-capacity Hopf bifurcation lies far above any real health system; the surveillance-delay bifurcation sits on the demographic timescale, so for an acute outbreak delay matters through overshoot and larger first waves, not sustained cycles.

Conclusion. Isolation capacity should be allocated by transmission structure and surveillance performance, not population. A modest, well-staffed isolation unit at Utange, paired with same-day testing on the Coast, would close most of Mombasa's gap.

Introduction

Bundibugyo virus (BDBV) is one of six orthoebolaviruses and one of four that cause disease in humans. It was first recognised in Bundibugyo District, Uganda, in 2007, and reappeared in Isiro, Democratic Republic of the Congo (DRC), in 2012. Both earlier outbreaks were small and well below the scale needed to estimate most transmission parameters. In May 2026 the DRC declared a BDBV outbreak in Ituri Province that became the largest Ebola outbreak recorded in the country; by 6 October 2026 it had caused 8,728 confirmed cases and 4,205 deaths, a crude case fatality of about 48% (WHO DON619). There is no licensed vaccine or specific therapeutic for Bundibugyo, so control rests on surveillance, isolation, contact tracing and safe and dignified burial.

On 6 October 2026 Kenya confirmed its first imported BDBV case: a Kenyan who had been living in the DRC, who travelled by road to Kampala and flew to Nairobi on 3 October, was isolated at a Nairobi hospital, and died. A second case, a quarantined family contact, was confirmed on 14 October. Contact tracing listed 128 contacts, of whom 108 were in quarantine in Kenya (WHO AFRO; Ministry of Health, via Citizen Digital). Kenya activated a national Ebola Incident Management System through the Kenya National Public Health Institute (KNPHI) in May 2026 and, by mid-October, reported screening more than 650,000 travellers and training nearly 5,000 health workers.

Kenya's preparedness has a structural weakness that modelling can make precise: isolation capacity is concentrated in the capital. Of roughly 241 designated isolation beds, the Ministry of Health lists 8 at Kenyatta National Hospital, 49 at the National Police Hospital and 130 at the (private) Nairobi Hospital, with 50 at Moi Teaching and Referral Hospital in Eldoret and just **four at Port Reitz Hospital in Mombasa** (Citizen Digital). Nairobi holds about 78% of the designated beds; Mombasa, Kenya's second city and principal seaport, with 1.2 million residents and road, rail and sea links running directly to the Great Lakes region, holds under 2%. Utange Field Hospital in Mombasa is also designated but was assigned no bed count.

Two features make Mombasa a distinct modelling problem rather than a scaled-down Nairobi. First, a larger share of its population lives in informal settlements, where crowding and shared sanitation raise contact rates. Second, Mombasa has no designated Ebola testing laboratory; samples travel to Nairobi, lengthening the time from symptom onset to isolation. Both act through the same mechanism: when sick people cannot be isolated promptly — because wards are full or because detection is slow — they keep transmitting in the community.

This study asks three questions. (1) If two imported cases seed an outbreak in each city, how likely is it to become self-sustaining? (2) How many isolation beds would Mombasa need, and specifically how large should an isolation unit at Utange be, given the four beds already at Port Reitz? (3) How do these requirements depend on surveillance speed? We answer them with a transmission model that treats isolation beds as a hard constraint, estimate the bed requirement stochastically, and use two bifurcation analyses to characterise the tipping behaviour that makes bed capacity and detection delay decisive. We frame surveillance delay through the WHO Public Health Intelligence (PHI) competency framework, whose Detect–Verify–Analyse–Assess sequence is the operational content of the delay parameter (WHO, 2025).

No previous Ebola modelling study, to our knowledge, has applied either a hospital-bed bifurcation analysis or a surveillance-delay bifurcation analysis to the disease; both are established in the mathematical-epidemiology literature for other settings. This article is a first draft intended to be refined as Kenyan surveillance and capacity data become available.

Methods

Study setting and populations

We model Nairobi City County and Mombasa County as separate closed populations, using the 2019 Kenya Population and Housing Census. Nairobi has 4,397,073 residents at 6,247 per km² with a mean household size of 2.9; Mombasa has 1,208,333 residents at 5,495 per km² with a household size of 3.1 (KNBS 2019). Each population is divided into a formal-housing group and an informal-settlement group. We set the informal-settlement share to 50% for Nairobi and 60% for Mombasa, within the ranges reported in World Bank and UN-Habitat sources, and assign informal-settlement residents a contact rate 1.6 times that of formal residents to represent crowding and shared sanitation. These two parameters are varied in sensitivity analysis.

For isolation capacity we use the Ministry of Health's October 2026 designations: Nairobi is modelled with 57 public-sector beds (Kenyatta National Hospital and the National Police Hospital, excluding the 130 private Nairobi Hospital beds), and Mombasa with the four beds at Port Reitz plus a variable number at Utange, which is the quantity we optimise. We do not reuse the COVID-era isolation figures, which overstate Ebola-grade capacity.

Transmission model

We use a deterministic SEIHFR compartmental model after Legrand et al. (2007), extended to two mixing groups and a bed-limited isolation rate. Within each group g (formal, informal) individuals are susceptible (S), exposed but not yet infectious (E), infectious in the community (I), isolated in hospital (H), dead but not yet buried (F), or removed by recovery or safe burial (R). Transmission occurs in the community, in hospital, and at unsafe burials, with coefficients β_I, β_H and β_F.

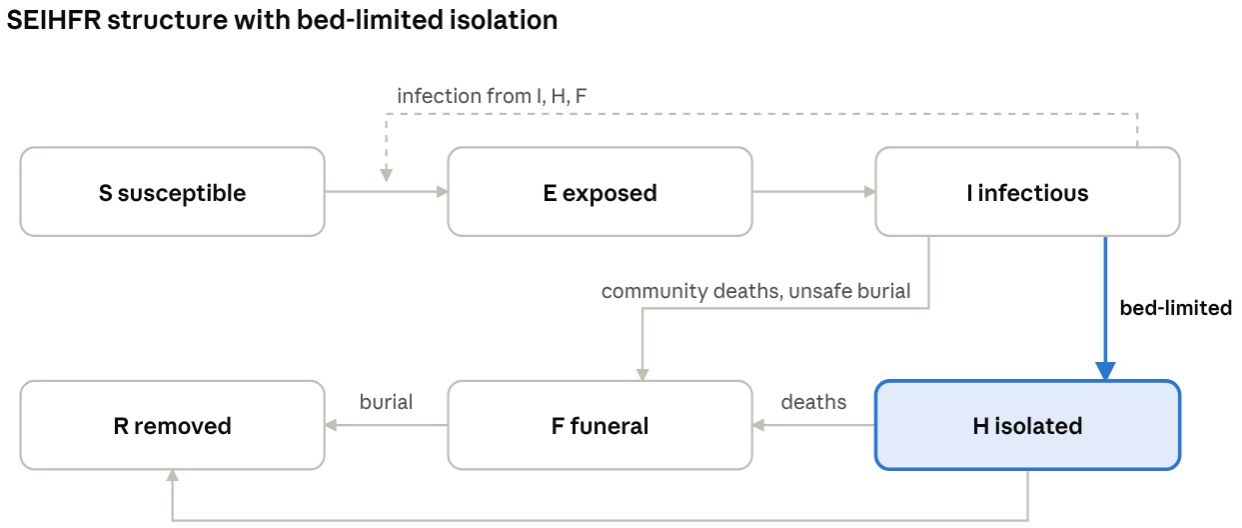


Figure 1. Model compartments and transmission routes, run separately for the formal and informal-settlement groups

The central mechanism is that admission to isolation saturates as beds fill. Let φ(H) be the fraction of normal admission still possible when H of B beds are occupied, a smoothed form of max(0, 1 − H/B). A symptomatic person who cannot be admitted remains in compartment I and keeps transmitting. The force of infection on group g, where c_g is the group's relative contact rate and N the population, is

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\lambda_g = c_g(1-u)\,\beta_I\frac{\sum_k c_k I_k}{\sum_k c_k N_k} +
c_g\,\beta_F\frac{\sum_k c_k F_k}{\sum_k c_k N_k} + \beta_H\frac{\sum_k H_k}
{N}
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and the admission flux from I to H is

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\text{admissions}_g = \rho_g\big[\varphi(H) + (1-\varphi(H))\,\omega\big] I_g
,
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where ρ_g is the baseline isolation rate (the reciprocal of the onset-to-isolation delay), u is the community-contact reduction achieved by the response, and ω is the fraction of overflow absorbed by community care centres or supported home isolation. The

remaining transitions follow the natural history: E→I at rate σ, I→ death or recovery at rate γ_I, H→ death or discharge at rate γ_H, and F→R (burial) at rate γ_F, with case fatality applied at the I and H exits and safe-burial fractions diverting deaths away from F.

Reproduction number. We compute the basic reproduction number R_0 as the spectral radius of the next-generation matrix $K = FV^{-1}$ built from the E, I, H, F subsystem of both groups, following van den Driessche and Watmough (2002) and Diekmann and Heesterbeek (2000). The same construction with the admission rate set to zero gives the reproduction number that applies once wards are full. We split R_0 into community, hospital and funeral contributions by zeroing the other two coefficients in turn.

Parameters and data sources

Bundibugyo-specific estimates exist for few parameters, so we take natural-history delays from Legrand et al. (2007) and transmissibility from the Bundibugyo and wider Ebola literature, labelling each substitution. The reference R_0 of 2.0 sits between the Uganda 2007 reanalysis ($R \approx 1.55$; de Padua & Akhmetzhanov 2026) and 2026 estimates for the DRC outbreak (phylodynamic $R \approx 1.95$ pre-decline; CDC scenario $R_0 \approx 2.51$); we report R_0 = 1.5, 2.0 and 2.5 as a sensitivity band but use 2.0 as the planning default.

Parameter	Value	Source
Incubation period (1/σ)	7 days	Bundibugyo, Uganda 2007 (Wamala 2010)
Onset to death or recovery, community (1/γ_I)	10 days	Legrand 2007
Isolation to death or discharge (1/γ_H)	6 days	Legrand 2007
Death to burial (1/γ_F)	2 days	Legrand 2007
Case fatality, isolated / community	31% / 47%	Uganda 2007; WHO DRC 2026
Reference R_0 (community : hospital : funeral)	2.0 (0.8 : 0.4 : 0.6)	Legrand 2007 via Gomes 2014
Onset to isolation, formal / informal	Nairobi 4 / 6 d; Mombasa 5 / 7 d	Assumption (no Coast lab)
Informal-settlement share	Nairobi 0.50; Mombasa 0.60	World Bank; UN-Habitat
Informal contact multiplier (c)	1.6	Assumption

Parameter	Value	Source
Overdispersion (negative-binomial k)	0.18 (0.37, 1.0)	Ebola, Althaus 2015
Safe burial, isolation / community deaths	90% / 50% rising to 90%	Assumption; Tiffany 2017
Response: community-contact reduction (υ)	60%	Scenario
Response start	7 days after first admission	Scenario (varied 3–21 d)

The transmission coefficients β_I, β_H and β_F are calibrated so that a homogeneous reference population with unlimited beds and a 5-day onset-to-isolation delay has the chosen R_0 split in Legrand's community : hospital : funeral proportions; each city's own R_0 then follows from its structure. Code, parameter files and an interactive version accompany this manuscript.

Stochastic model, outbreak probability and bed sizing

Because an importation involves small numbers, we complement the deterministic model with a stochastic treatment. We first estimate the probability that two imported cases start a self-sustaining outbreak using a branching process with a negative-binomial offspring distribution, mean equal to the effective reproduction number and dispersion k. For a mean R and dispersion k the per-lineage extinction probability q is the smallest root in [0,1] of

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q = \left(\frac{k}{k + R(1-q)}\right)^{k}, \qquad P(\text{outbreak}\mid
n\text{ cases}) = 1 - q^{n},
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(Lloyd-Smith et al. 2005). We take n = 2 and report k = 0.18, 0.37 and 1.0.

To size isolation capacity we run the full two-group SEIHFR as a stochastic tau-leaping process (binomial increments, 0.25-day step) over many realisations, seeding two infectious cases in the informal-settlement group. Admission is a hard constraint: a case is admitted only if a bed is free, otherwise it stays infectious in the community, which is the stochastic analogue of the saturating admission rate and matches the bed-dependent admission of Kucharski et al. (2015). The response begins a fixed delay τ after the first admission (our proxy for detection), cutting community contacts by υ, speeding isolation and raising safe-burial coverage. For each candidate bed number we estimate the probability that cumulative cases stay below 20,000 over 400 days ("containment") and

the distribution of peak occupancy. The required bed number is the smallest total that reaches the target containment probability; subtracting the four Port Reitz beds gives the Utange requirement. We repeat the search for τ = 3, 7, 14 and 21 days.

Bifurcation analyses

We characterise the tipping behaviour of the system with two bifurcation analyses that connect to established theory.

Bed capacity. A plain outbreak model has no endemic equilibrium and so cannot show a Hopf bifurcation. Following the hospital-bed literature (Wang and Ruan 2004; Wan and Cui 2013; Shan and Zhu 2014), we add demographic turnover (births and non-Ebola deaths at rate μ, with life expectancy 64 years) and a saturating admission rate $\rho B/(B+H)$. We trace the endemic equilibrium as bed capacity B falls and compute the eigenvalues of the Jacobian along the branch, locating Hopf points where a complex conjugate pair crosses the imaginary axis and the backward-bifurcation fold where a real eigenvalue does. Bed capacity is expressed per 100,000 people so the two cities are comparable.

Surveillance delay. We then let the response itself lag the epidemic. The community-contact reduction at time t depends on reported prevalence at time $t - \tau$, where τ is the surveillance-to-response delay — operationally the Detect–Verify–Analyse–Assess chain of the WHO PHI framework plus the time to act. This is an information-dependent control with delay, in the family of Sirijampa et al. (2018) and d'Onofrio and Manfredi. We integrate the resulting delay-differential system for a range of τ and locate the critical delay τ_0 at which post-transient oscillations change from damped to growing, i.e. the Hopf threshold. Because this analysis is set at the endemic equilibrium, we interpret τ_0 for the acute outbreak as indicating the regime in which delayed feedback produces overshoot and resurgence rather than a literal sustained cycle, consistent with the long critical delays reported in that literature.

Results

Reproduction numbers and infection rate

At a reference R_0 of 2.0, the structured model gives R_0 = 1.95 for Nairobi and 2.18 for Mombasa while isolation beds are available. Mombasa's higher value follows from its larger informal-settlement share and longer onset-to-isolation delay. The calibrated transmission coefficients are β_I = 0.267, β_H = 0.111 and β_F = 0.917 per day. The implied early epidemic growth rate is 0.065 per day in Nairobi (doubling time 10.6 days) and 0.077 per day in Mombasa (doubling time 9.0 days), bracketing the 9–14 day doubling times reported for the 2026 DRC outbreak.

The decisive quantity is what happens when wards fill. With admission blocked, each community case transmits for about ten days rather than three, and R_0 rises to 4.24 in Nairobi and 4.41 in Mombasa — more than double. Community transmission supplies 1.28 of Nairobi's R_0 and 1.50 of Mombasa's; hospital and funeral settings contribute 0.41–0.45 and 0.26–0.29 respectively. The gap between the bed-available and ward-full reproduction numbers is the quantity the isolation system exists to prevent.

Outbreak probability from two index cases

Ebola transmission is strongly overdispersed: a minority of cases cause most onward infection, so many introductions die out by chance even when the reproduction number exceeds one. With two index cases and the Ebola dispersion estimate k = 0.18 (Althaus 2015), the probability of a self-sustaining outbreak is 32% in Nairobi and 36% in Mombasa. If transmission were less overdispersed the probability would be much higher — 50–55% at k = 0.37 and 74–79% with no overdispersion (k = 1.0).

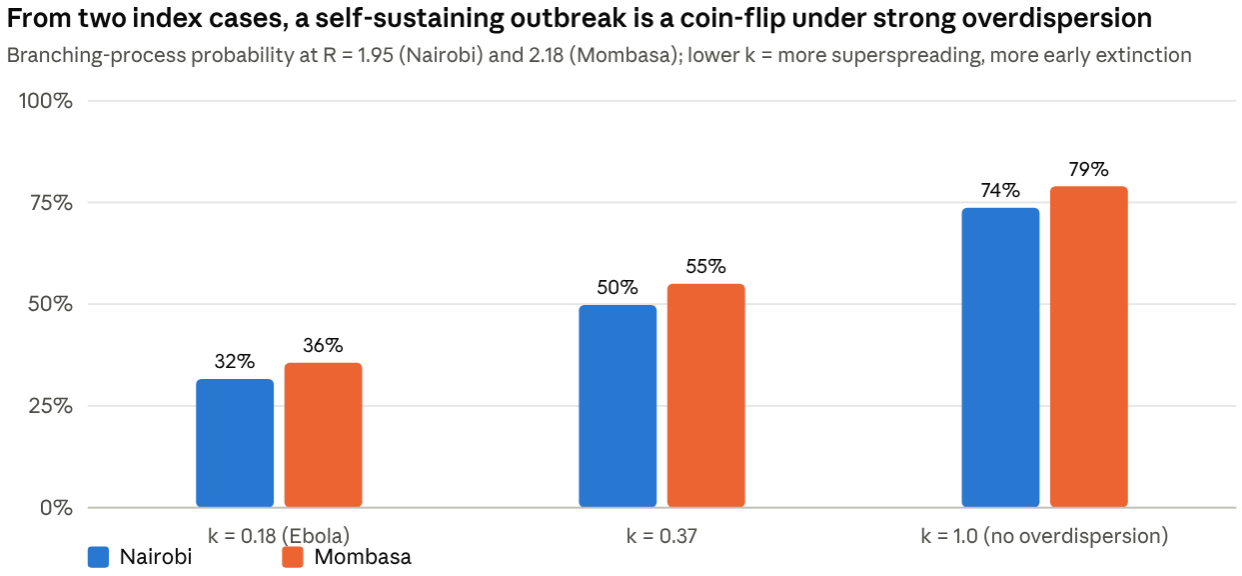


Figure 2. Probability that two imported cases start a self-sustaining outbreak, by dispersion k

Two points follow. First, roughly two-thirds of introductions at this dispersion fizzle out before establishing, which is consistent with the single imported chain Kenya has so far contained. Second, the one-third that do establish are the reason capacity must be planned for the tail, not the average: the same overdispersion that makes extinction likely also produces occasional explosive superspreading. The bed-sizing below is therefore conditioned on an outbreak having taken off.

Containment probability and the Utange bed requirement

Conditional on an outbreak taking off, the probability that it stays contained rises steeply with isolation beds and then saturates. With the response beginning seven days after the first admission and a 60% cut in community contacts, Mombasa's four existing Port Reitz beds give only a 56% chance of containment. Containment reaches 90% at about 16 total beds, 95% at about 22, and 99% at about 36.

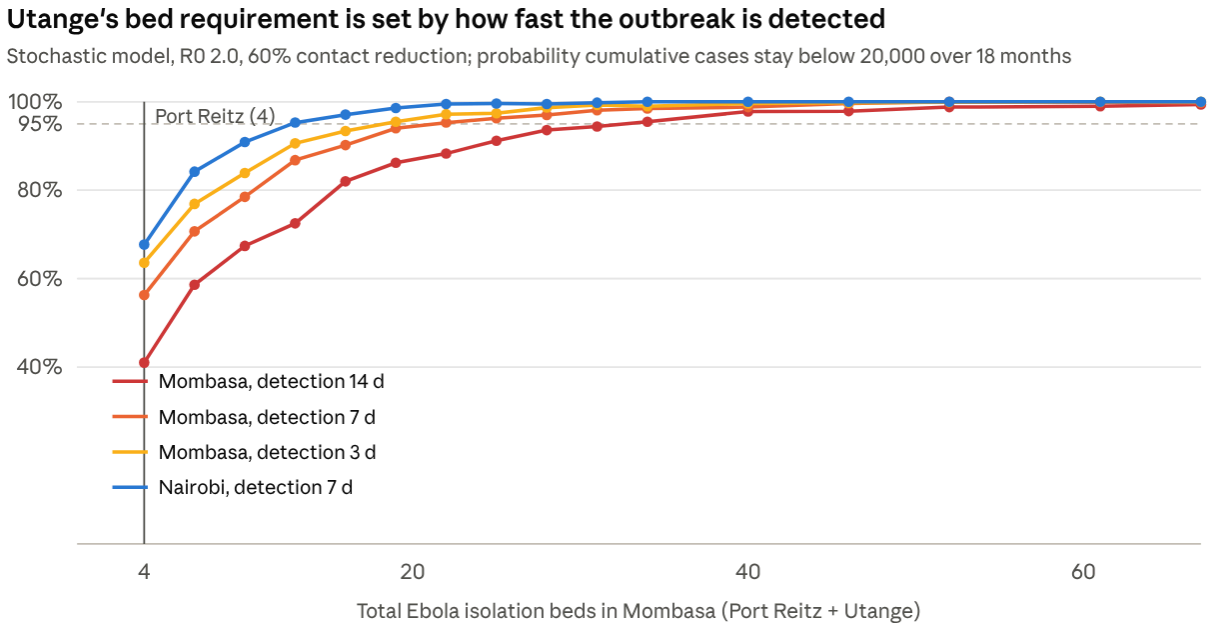


Figure 3. Probability of containment versus total isolation beds in Mombasa, at three detection delays, with Nairobi for comparison

Subtracting the four Port Reitz beds, **Utange would need about 18 isolation beds for a 95% probability of containment and about 32 for 99%.** Nairobi's 57 public-sector beds, by contrast, deliver near-certain containment at every detection delay we tested, confirming that the capital is well provisioned and Mombasa is the binding gap. The requirement is modest in absolute terms — an isolation unit of a few dozen beds — but it is roughly five times the capacity Mombasa currently has designated.

Surveillance delay and the bed tradeoff

How fast the outbreak is detected is as decisive as how many beds exist, because beds and detection substitute for each other. The minimum total beds Mombasa needs for a 95% probability of containment rises from about 18 when the response begins three days after the first admission, to 22 at seven days, to 32 at fourteen days, to 50 at twenty-one days. In Utange terms, the isolation unit would need roughly 14, 18, 28 or 46 beds across that range of detection speeds. The 99% requirement rises even more steeply, from 34 total beds at three days to 76 at twenty-one.

Detection delay after first admission	Total beds for 95%	Utange beds for 95%	Total beds for 99%
3 days	18	14	34
7 days	22	18	36
14 days	32	28	50
21 days	50	46	76

The policy reading is direct: investment in faster detection on the Coast — same-day testing, lower reporting thresholds, active case-finding in the high-risk sub-counties — substitutes for physical beds. Halving the detection delay from 14 to 7 days cuts the 95% bed requirement by about a third. A county laboratory or a deployed mobile laboratory at Coast General or Port Reitz, which removes the day samples currently spend travelling to Nairobi, is likely cheaper than the beds it saves.

Bifurcation analysis

Bed capacity. The long-run model has a Hopf bifurcation in bed capacity, but at roughly 2,800–3,500 isolation beds per 100,000 people — more than twenty times Kenya's total hospital-bed density of about 120 per 100,000, and far beyond any plausible Ebola isolation capacity. Above the Hopf point the endemic equilibrium is stable; between about 60 and 2,800 beds per 100,000 the model produces recurrent explosive waves separated by near-extinction; below about 60 the equilibrium repels without oscillation. At low reference R_0 a backward bifurcation appears, so that a reproduction number below one does not by itself guarantee elimination when beds are scarce.

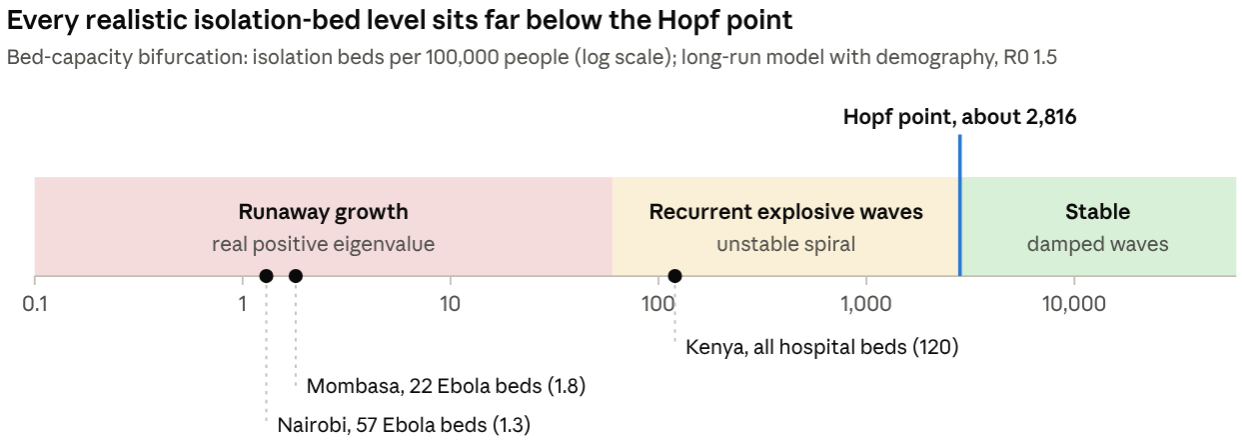


Figure 4. Bed-capacity bifurcation: long-run regimes by isolation beds per 100,000, with real health-system levels marked

The practical reading is that Ebola-type diseases sit in the unstable region of this diagram in any real health system, which is why they appear as outbreaks rather than as endemic disease. The bifurcation is mathematically real but not a usable planning target; the operational threshold is the outbreak-scale tipping point in the containment analysis above, at tens of beds per city.

Surveillance delay. In the delayed-response model the endemic equilibrium also loses stability through a Hopf bifurcation as the response delay grows, but the critical delay is about 1,170 days. This long value is expected: the oscillation period is set by the slow demographic replenishment of susceptibles, and published critical delays in information-feedback models are similarly on the scale of months to years. The honest interpretation for an acute Ebola outbreak is therefore not that a delay of this length produces a sustained cycle, but that delayed control acts through overshoot — a response that lags the epidemic lets the first wave grow larger before it turns over, and can permit resurgence. That overshoot is exactly what the stochastic bed-versus-delay result above quantifies: the same outbreak needs far more isolation capacity when the response is late.

Discussion

Three findings have direct operational implications for Kenya's public hospitals.

Allocate isolation capacity by transmission risk, not population. Mombasa needs more isolation beds than its population share would suggest. A per-capita split of a national isolation budget would give Mombasa about a quarter of Nairobi's beds; our model implies it needs an isolation unit of a few dozen, against the four currently designated, while Nairobi's 57 public beds are already sufficient. The gap is structural, driven by Mombasa's larger informal-settlement population and its longer path to a test result. County isolation targets should be set from the tipping analysis: at a planning R_0 of 2.0 and a realistic community response, on the order of 20–40 Ebola-grade beds for Mombasa with surge plans to roughly double them, concentrated at Utange as the county isolation centre with Port Reitz as the first-line unit.

Detection speed is interchangeable with beds, and usually cheaper. Because a late response forces more cases through the community, the bed requirement roughly triples as detection slows from a few days to three weeks. The WHO Public Health Intelligence framework gives the operational content of that delay: the Detect, Verify, Analyse and Assess steps each add days, and each is a target for compression. The single most valuable investment the model identifies is same-day testing on the Coast. Mombasa currently has no designated Ebola laboratory, so a suspected case waits for transport to

Nairobi; a county or deployed mobile laboratory removes that delay and, in the model, cuts the bed requirement by roughly a third. Active case-finding and lower alert thresholds in Kisauni, Likoni, Changamwe and Jomvu act the same way.

Plan community care and safe burial as distinct resource lines. Isolation beds are not the only sink for cases. In the 2026 DRC outbreak about 71% of deaths occurred in the community rather than in treatment centres, so bed demand scales with detected cases, not with incidence; capacity sized from projected deaths would understate the need for community case-finding, safe and dignified burial, and transport. Our scenarios show that absorbing even a third of overflow into community care centres or supported home isolation materially lowers the bed requirement, and that keeping safe-burial coverage high suppresses the funeral transmission route, which contributes a quarter to a third of the reproduction number. Burial teams that respect Coastal practice, including rapid burial customs, are a specific need distinct from beds.

The bifurcation results frame why these levers matter. The bed-capacity analysis shows that an isolation system operating near capacity sits close to a tipping point: once community prevalence exceeds the beds available, the effective reproduction number more than doubles and the outbreak can jump to a high-prevalence state. The surveillance-delay analysis shows that control which lags the epidemic overshoots. Neither is a precise forecast, but together they explain why Mombasa's thin capacity and slower detection are not a modest disadvantage but a qualitative vulnerability.

This study is, to our knowledge, the first to apply hospital-bed and surveillance-delay bifurcation analyses to Ebola, and the first to size an Ebola isolation unit from the quantiles of a stochastic occupancy simulation. The methods transfer directly to other high-consequence pathogens and other counties.

Limitations

The results are planning scenarios, and four assumptions have no direct Kenyan data behind them: the informal-settlement contact multiplier, the onset-to-isolation delays, the effect size of the response package, and the isolation-bed designations. The bed thresholds should be read as orders of magnitude.

- **Parameter voids.** Bundibugyo has been characterised in only two earlier outbreaks. Transmissibility is now triangulated from the 2026 outbreak, but the length of isolation stay — the parameter that most directly drives bed-days — has no usable published distribution for Bundibugyo, and survivors occupy beds longer than fatal cases, so a falling case fatality would raise mean occupancy. We take delays from Zaire-ebolavirus studies and label the substitution; these values should be updated as 2026 estimates appear.

- **Deterministic and stochastic scope.** The deterministic model overstates spread in the first weeks, which the stochastic branching process corrects for establishment but not for the full peak-occupancy distribution. We did not fit to Kenyan case data, which do not yet exist at sufficient scale.
- **No travel coupling.** Nairobi and Mombasa are modelled as closed. They are linked by road, rail and air, so an outbreak in one would seed the other; a metapopulation version with Northern Corridor mobility is the natural next step.
- **Intervention effect sizes are borrowed.** All come from Zaire or Sudan ebolavirus outbreaks, not Bundibugyo, and from different health systems. The contact-reduction, safe-burial and community-care effects are plausible but not Kenya-specific.
- **Structural omissions.** The model has no explicit health-worker compartment, no age structure (case fatality is highest in young children), and no vaccine, for which none exists for Bundibugyo. The surveillance-delay analysis uses a single lumped delay rather than the full Detect–Verify–Analyse–Assess chain as separate stages.

These are the priorities for the next iteration: a stochastic estimate of the probability that a single importation establishes, an explicit length-of-stay distribution split by outcome, Nairobi–Mombasa travel coupling, a staged surveillance-delay model mapped to the PHI framework, and calibration to Kenyan contact-tracing data from the October 2026 cases if the Ministry of Health releases it.

Conclusion

Kenya's first imported Bundibugyo cases arrived into a system whose isolation capacity is concentrated in Nairobi. Our model suggests the capital is well provisioned and that the binding gap is on the Coast, where Mombasa has four designated beds against a modelled need of a few dozen. The requirement is modest in absolute terms and falls sharply if detection is fast: an isolation unit of roughly 18–30 beds at Utange, paired with same-day testing in Mombasa and prepared community-care and safe-burial capacity, would close most of the gap. Allocating isolation capacity by transmission risk and surveillance performance, rather than by population, is the central recommendation. These are first-draft figures; the framework is built to be sharpened as Kenyan parameters and capacity data become available.

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A companion literature synthesis with the full evidence base and additional citations accompanies this manuscript.